

AC ELECTRICAL SPECIFICATIONS — READ AND WRITE CYCLES

(VCC = 5.0 Vdc $\pm$ 5%; GND = 0 Vdc; 40 MHz-T_A = 0° to 70°C, 50 MHz-T_A = 0°C to T_C = 80°C)
(see Figures 3–8)

Num.	Characteristic	20 MHz		25 MHz		33.33 MHz		40 MHz		50 MHz*		Unit
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
6	Clock High to Function Code, Size, RMC, IPEND, CIOUT, Address Valid	0	25	0	20	0	14	0	14	0	14	ns
6A	Clock High to $\overline{\text{ECS}}$, $\overline{\text{OCS}}$ Asserted	0	15	0	15	0	12	0	10	0	10	ns
6B	Function Code, Size, RMC, IPEND, CIOUT, Address Valid to Negating Edge of $\overline{\text{ECS}}$	4	—	3	—	3	—	3	—	3	—	ns
7	Clock High to Function Code, Size, RMC, CIOUT, Address, Data High Impedance	0	50	0	40	0	30	0	25	0	20	ns
8	Clock High to Function Code, Size, RMC, IPEND, CIOUT, Address Invalid	0	—	0	—	0	—	0	—	0	—	ns
9	Clock Low to $\overline{\text{AS}}$, $\overline{\text{DS}}$ Asserted, $\overline{\text{CBREQ}}$ Valid	3	20	3	18	2	10	2	10	2	10	ns
9A ¹	$\overline{\text{AS}}$ to $\overline{\text{DS}}$ Assertion Skew (Read)	–10	10	–10	10	–8	8	–6	6	–6	6	ns
9B ¹⁴	$\overline{\text{AS}}$ Asserted to $\overline{\text{DS}}$ Asserted (Write)	32	—	27	—	22	—	16	—	14	—	ns
10	$\overline{\text{ECS}}$ Width Asserted	15	—	10	—	8	—	5	—	4	—	ns
10A	$\overline{\text{OCS}}$ Width Asserted	15	—	10	—	8	—	5	—	4	—	ns
10B ⁷	$\overline{\text{ECS}}$, $\overline{\text{OCS}}$ Width Negated	10	—	5	—	5	—	5	—	4	—	ns
11	Function Code, Size, RMC, CIOUT, Address Valid to Asserting Edge of $\overline{\text{AS}}$ Asserted (and $\overline{\text{DS}}$ Asserted, Read)	10	—	7	—	5	—	5	—	3	—	ns
12	Clock Low to $\overline{\text{AS}}$, $\overline{\text{DS}}$, $\overline{\text{CBREQ}}$ Negated	0	20	0	18	0	10	0	10	0	10	ns
12A	Clock Low to $\overline{\text{ECS}}$ / $\overline{\text{OCS}}$ Negated	0	20	0	18	0	15	0	12	0	11	ns
13	$\overline{\text{AS}}$, $\overline{\text{DS}}$ Negated to Function Code, Size, RMC, CIOUT, Address Invalid	10	—	7	—	5	—	3	—	3	—	ns
14	$\overline{\text{AS}}$ (and $\overline{\text{DS}}$ Read) Width Asserted (Asynchronous Cycle)	85	—	70	—	45	—	30	—	25	—	ns
14A ¹¹	$\overline{\text{DS}}$ Width Asserted (Write)	38	—	30	—	23	—	18	—	13	—	ns
14B	$\overline{\text{AS}}$ (and $\overline{\text{DS}}$, Read) Width Asserted (Synchronous Cycle)	35	—	30	—	23	—	18	—	13	—	ns
15	$\overline{\text{AS}}$, $\overline{\text{DS}}$ Width Negated	38	—	30	—	23	—	18	—	13	—	ns
15A ⁸	$\overline{\text{DS}}$ Negated to $\overline{\text{AS}}$ Asserted	30	—	25	—	18	—	16	—	14	—	ns
16	Clock High to $\overline{\text{AS}}$, $\overline{\text{DS}}$, R/W, $\overline{\text{DBEN}}$, $\overline{\text{CBREQ}}$ High Impedance	—	50	—	40	—	30	—	25	—	20	ns

AC ELECTRICAL SPECIFICATIONS (Continued)

Num.	Characteristic	20 MHz		25 MHz		33.33 MHz		40 MHz		50 MHz*		Unit
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
17	$\overline{AS}$, $\overline{DS}$ Negated to $R/\overline{W}$ Invalid	10	—	7	—	5	—	3	—	3	—	ns
18	Clock High to $R/\overline{W}$ High	0	25	0	20	0	15	0	14	0	14	ns
20	Clock High to $R/\overline{W}$ Low	0	25	0	20	0	15	0	14	0	14	ns
21	$R/\overline{W}$ High to $\overline{AS}$ Asserted	10	—	7	—	5	—	5	—	3	—	ns
22	$R/\overline{W}$ Low to $\overline{DS}$ Asserted (Write)	60	—	47	—	35	—	24	—	23	—	ns
23	Clock High to Data-Out Valid	—	25	—	20	—	14	—	14	—	14	ns
24	Data-Out Valid to Negating Edge of $\overline{AS}$	8	—	5	—	3	—	3	—	3	—	ns
25 ¹¹	$\overline{AS}$, $\overline{DS}$ Negated to Data-Out Invalid	10	—	7	—	5	—	3	—	3	—	ns
25A ^{9,11}	$\overline{DS}$ Negated to $\overline{DBEN}$ Negated (Write)	10	—	7	—	5	—	3	—	3	—	ns
26 ¹¹	Data-Out Valid to Asserting Edge of $\overline{DS}$ Asserted (Write)	10	—	7	—	5	—	3	—	3	—	ns
27	Data-In Valid to Clock Low (Setup)	4	—	2	—	1	—	1	—	1	—	ns
27A	Late $\overline{BERR}/\overline{HALT}$ Asserted to Clock Low (Setup)	10	—	5	—	3	—	3	—	3	—	ns
28 ¹²	$\overline{AS}$, $\overline{DS}$ Negated to $\overline{DSACKx}$, $\overline{BERR}$, $\overline{HALT}$, $\overline{AVEC}$ Negated (Asynchronous Hold)	0	50	0	40	0	30	0	20	0	15	ns
28A ¹²	Clock Low to $\overline{DSACKx}$, $\overline{BERR}$, $\overline{HALT}$, $\overline{AVEC}$ Negated (Synchronous Hold)	12	85	8	70	6	50	6	40	6	35	ns
29 ¹²	$\overline{AS}$, $\overline{DS}$ Negated to Data-In Invalid (Asynchronous Hold)	0	—	0	—	0	—	0	—	0	—	ns
29A ¹²	$\overline{AS}$, $\overline{DS}$ Negated to Data-In High Impedance	—	50	—	40	—	30	—	25	—	20	ns
30 ¹²	Clock Low to Data-In Invalid (Synchronous Hold)	12	—	8	—	6	—	6	—	6	—	ns
30A ¹²	Clock Low to Data-In High Impedance (Read followed by Write)	—	75	—	60	—	45	—	30	—	25	ns
31 ²	$\overline{DSACKx}$ Asserted to Data-In Valid (Asynchronous Data Setup)	—	43	—	28	—	20	—	14	—	13	ns
31A ³	$\overline{DSACKx}$ Asserted to $\overline{DSACKx}$ Valid (Skew)	—	10	—	7	—	5	—	3	—	3	ns
32	RESET Input Transition Time	—	1.5	—	1.5	—	1.5	—	1.5	—	1.5	Clks
33	Clock Low to $\overline{BG}$ Asserted	0	25	0	20	0	15	0	14	0	14	ns
34	Clock Low to $\overline{BG}$ Negated	0	25	0	20	0	15	0	14	0	14	ns
35	$\overline{BR}$ Asserted to $\overline{BG}$ Asserted (RMC Not Asserted)	1.5	3.5	1.5	3.5	1.5	3.5	1.5	3.5	1.5	3.5	Clks

AC ELECTRICAL SPECIFICATIONS (Continued)

Num.	Characteristic	20 MHz		25 MHz		33.33 MHz		40 MHz		50 MHz*		Unit
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
37	$\overline{\text{BGACK}}$ Asserted to $\overline{\text{BG}}$ Negated	1.5	3.5	1.5	3.5	1.5	3.5	1.5	3.5	1.5	3.5	Clks
37A ⁶	$\overline{\text{BGACK}}$ Asserted to $\overline{\text{BR}}$ Negated	0	1.5	0	1.5	0	1.5	0	1.5	0	1.5	Clks
39	$\overline{\text{BG}}$ Width Negated	75	—	60	—	45	—	30	—	30	—	ns
39A	$\overline{\text{BG}}$ Width Asserted	75	—	60	—	45	—	30	—	30	—	ns
40 ¹	Clock High to $\overline{\text{DBEN}}$ Asserted (Read)	0	25	0	20	0	18	0	16	0	14	ns
41	Clock Low to $\overline{\text{DBEN}}$ Negated (Read)	0	25	0	20	0	18	0	16	0	14	ns
42	Clock Low to $\overline{\text{DBEN}}$ Asserted (Write)	0	25	0	20	0	18	0	16	0	14	ns
43	Clock High to $\overline{\text{DBEN}}$ Negated (Write)	0	25	0	20	0	18	0	16	0	14	ns
44	R/W Low to $\overline{\text{DBEN}}$ Asserted (Write)	10	—	7	—	5	—	5	—	5	—	ns
45 ⁵	$\overline{\text{DBEN}}$ Width Asserted Asynchronous Read Asynchronous Write	50 100	— —	40 80	— —	30 60	— —	22 45	— —	20 40	— —	ns
45A ⁹	$\overline{\text{DBEN}}$ Width Asserted Synchronous Read Synchronous Write	10 50	— —	5 40	— —	5 30	— —	5 22	— —	5 20	— —	ns
46	R/W Width Asserted (Asynchronous Write or Read)	125	—	100	—	75	—	50	—	40	—	ns
46A	R/W Width Asserted (Synchronous Write or Read)	75	—	60	—	45	—	30	—	25	—	ns
47A	Asynchronous Input Setup Time to Clock Low	4	—	2	—	2	—	2	—	2	—	ns
47B	Asynchronous Input Hold Time from Clock Low	12	—	8	—	6	—	6	—	6	—	ns
48 ⁴	$\overline{\text{DSACKx}}$ Asserted to $\overline{\text{BERR}}$, $\overline{\text{HALT}}$ Asserted	—	20	—	25	—	18	—	14	—	13	ns
53	Data-Out Hold from Clock High	3	—	3	—	2	—	2	—	2	—	ns
55	R/W Asserted to Data Bus Impedance Change	25	—	20	—	15	—	11	—	11	—	ns
56	RESET Pulse Width (Reset Instruction)	512	—	512	—	512	—	512	—	512	—	Clks
57	$\overline{\text{BERR}}$ Negated to $\overline{\text{HALT}}$ Negated (Rerun)	0	—	0	—	0	—	0	—	0	—	ns
58 ¹⁰	$\overline{\text{BGACK}}$ Negated to Bus Driven	1	—	1	—	1	—	1	—	1	—	Clks
59 ¹⁰	$\overline{\text{BG}}$ Negated to Bus Driven	1	—	1	—	1	—	1	—	1	—	Clks

AC ELECTRICAL SPECIFICATIONS (Concluded)

Num.	Characteristic	20 MHz		25 MHz		33.33 MHz		40 MHz		50 MHz*		Unit
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
60 ¹³	Synchronous Input Valid to Clock High (Setup Time)	4	—	2	—	2	—	2	—	2	—	ns
61 ¹³	Clock High to Synchronous Input Invalid (Hold Time)	12	—	8	—	6	—	6	—	6	—	ns
62	Clock Low to $\overline{\text{STATUS}}$, REFILL Asserted	0	25	0	20	0	15	0	15	0	15	ns
63	Clock Low to $\overline{\text{STATUS}}$, REFILL Negated	0	25	0	20	0	15	0	15	0	15	ns

NOTES:

*T_{case} = 80°C Maximum

1. This number can be reduced to 5 ns if strobes have equal loads.
2. If the asynchronous setup time (#47A) requirements are satisfied, the $\overline{\text{DSACKx}}$ low to data setup time (#31) and $\overline{\text{DSACKx}}$ low to $\overline{\text{BERR}}$ low setup time (#48) can be ignored. The data must only satisfy the data-in clock low setup time (#27) for the following clock cycle, and $\overline{\text{BERR}}$ must only satisfy the late $\overline{\text{BERR}}$ low to clock low setup time (#27A) for the following clock cycle.
3. This parameter specifies the maximum allowable skew between $\overline{\text{DSACK0}}$ to $\overline{\text{DSACK1}}$ asserted or $\overline{\text{DSACK1}}$ to $\overline{\text{DSACK0}}$ asserted; specification #47A must be met by $\overline{\text{DSACK0}}$ or $\overline{\text{DSACK1}}$.
4. This specification applies to the first ($\overline{\text{DSACK0}}$ or $\overline{\text{DSACK1}}$) $\overline{\text{DSACKx}}$ signal asserted. In the absence of $\overline{\text{DSACKx}}$, $\overline{\text{BERR}}$ is an asynchronous input using the asynchronous input setup time (#47A).
5. $\overline{\text{DBEN}}$ may stay asserted on consecutive write cycles.
6. The minimum values must be met to guarantee proper operation. If this maximum value is exceeded, $\overline{\text{BG}}$ may be reasserted.
7. This specification indicates the minimum high time for $\overline{\text{ECS}}$ and $\overline{\text{OCS}}$ in the event of an internal cache hit followed immediately by another cache hit, a cache miss, or an operand cycle.
8. This specification guarantees operation with the MC68881/MC68882, which specifies a minimum time for $\overline{\text{DS}}$ negated to $\overline{\text{AS}}$ asserted (specification #13A in the *MC68881/MC68882 User's Manual*). Without this specification, incorrect interpretation of specifications #9A and #15 would indicate that the MC68030 does not meet the MC68881/MC68882 requirements.
9. This specification allows a system designer to guarantee data hold times on the output side of data buffers that have output enable signals generated with $\overline{\text{DBEN}}$. The timing on $\overline{\text{DBEN}}$ precludes its use for synchronous READ cycles with no wait states.
10. These specifications allow system designers to guarantee that an alternate bus master has stopped driving the bus when the MC68030 regains control of the bus after an arbitration sequence.
11. $\overline{\text{DS}}$ will not be asserted for synchronous write cycles with no wait states.
12. These hold times are specified with respect to strobes (asynchronous) and with respect to the clock (synchronous). The designer is free to use either time.
13. Synchronous inputs must meet specifications #60 and #61 with stable logic levels for *all* rising edges of the clock while $\overline{\text{AS}}$ is asserted. These values are specified relative to the high level of the rising clock edge. The values originally published were specified relative to the low level of the rising clock edge.
14. This specification allows system designers to qualify the $\overline{\text{CS}}$ signal of an MC68881/MC68882 with $\overline{\text{AS}}$ (allowing 7 ns for a gate delay) and still meet the $\overline{\text{CS}}$ to $\overline{\text{DS}}$ setup time requirement (spec 8B) of the MC68881/MC68882.